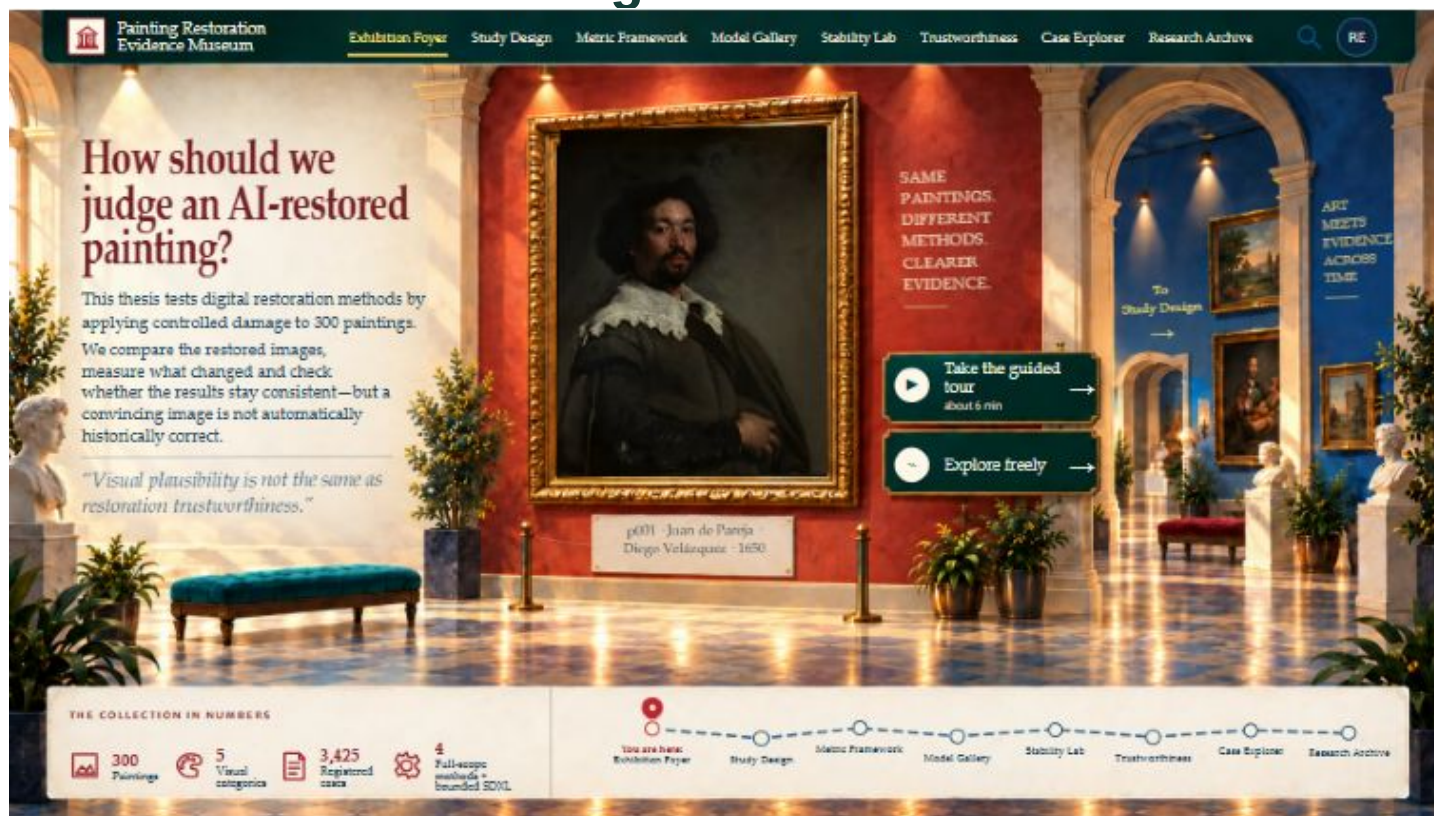


Supervisor dashboard walkthrough

Trustworthy Evaluation Frameworks for AI-Assisted Painting Restoration



EIGHT ROOMS. ONE EVIDENCE TRAIL.

A visual presentation guide to the completed study, built from the live deployed dashboard.

300 paintings

4 full-coverage methods

11 quality anchors

1,025 four-seed groups

Live screenshots + enlarged details + presentation notes + quantitative findings

[Open the Controlled-300 dashboard](#)

Choose a six-minute route

Page pairs combine a live room view with a closer reading. Click a row or use PDF bookmarks.

Exhibition Foyer	The question and the guided route	0:00-0:30	5
Study Design	Population, controls and experimental branches	0:30-1:15	7
Metric Framework	Why one similarity score is not enough	1:15-2:00	9
Model Gallery	The main method comparison	2:00-2:45	11
Stability Lab	Damage, placement and repeated outputs	2:45-3:30	13
Trustworthiness	One traceable review flag	3:30-4:15	15
Case Explorer	One candidate, its images and saved values	4:15-5:15	17
Research Archive	Reports, lineage and delivery	5:15-6:00	19

Before the tour: research answers (p. 3), scope and pipeline (p. 4).
Optional extension: D02 portrait review (pp. 21-22), about 2 extra minutes.
Reference: interpretation rules (p. 23) and evidence links (p. 24).

HOW TO USE THIS GUIDE

The timing is a suggested presentation script, not measured navigation performance. Screenshots show the loaded default rooms on 02 October 2026; controls were not exhaustively exercised for this guide.

Three questions. Three measured answers.

RQ1 / WHAT DOES RICHER EVALUATION REVEAL?

It separates structural resemblance from perceptual, feature and local consistency. Telea leads crop SSIM (0.8632 vs LaMa 0.8577), but LaMa has lower LPIPS (0.1292 vs 0.2007), lower masked colour error (6.88 vs 7.77) and higher DINOv2 similarity (0.8712 vs 0.6870).

RQ2 / HOW DO METHODS DIFFER?

LaMa leads 10 of 11 overall quality anchors. Mean masked MAE is 14.83 for LaMa, 17.57 for Telea, 19.16 for HINT and 30.61 for Stable Diffusion. All 11 canonical painting-level comparisons distinguish the methods (300 paintings; adjusted q below 10^{-134}). LaMa also has lower mask-placement dispersion than Stable Diffusion.

RQ3 / WHAT DOES REPEATED-OUTPUT DISAGREEMENT ADD?

The 1,025 four-seed groups expose spatial and scalar instability. At painting level, generic-prompt RGB variability correlates with masked MAE (Spearman ρ 0.513), and LPIPS disagreement with reference LPIPS error (0.586). Scratch-aware values are 0.622 and 0.713. Variability tracks different errors depending on the diagnostic.

Evidence trail N33 tables t04, t07 and t10; N26 statistical_results.csv, uncertainty_performance_association. Overall means exclude zero controls.

From paintings to traceable conclusions

Keep paintings, cases, candidates and repeated-seed groups distinct throughout the presentation.

Unit	Final scope	What the number means
Paintings	300	60 in each of five operational visual categories
Registered cases	3,425	Canonical and three controlled extension branches
Eligible restoration cases	2,620	2,320 nonzero damage + 300 identity controls
Matched primary candidates	10,480	One primary candidate per eligible case and full method
Retained catalogue	13,879	Includes repeated-seed, prompt and bounded SDXL evidence
Four-seed groups	1,025	780 canonical prompt-specific + 245 damage-size groups

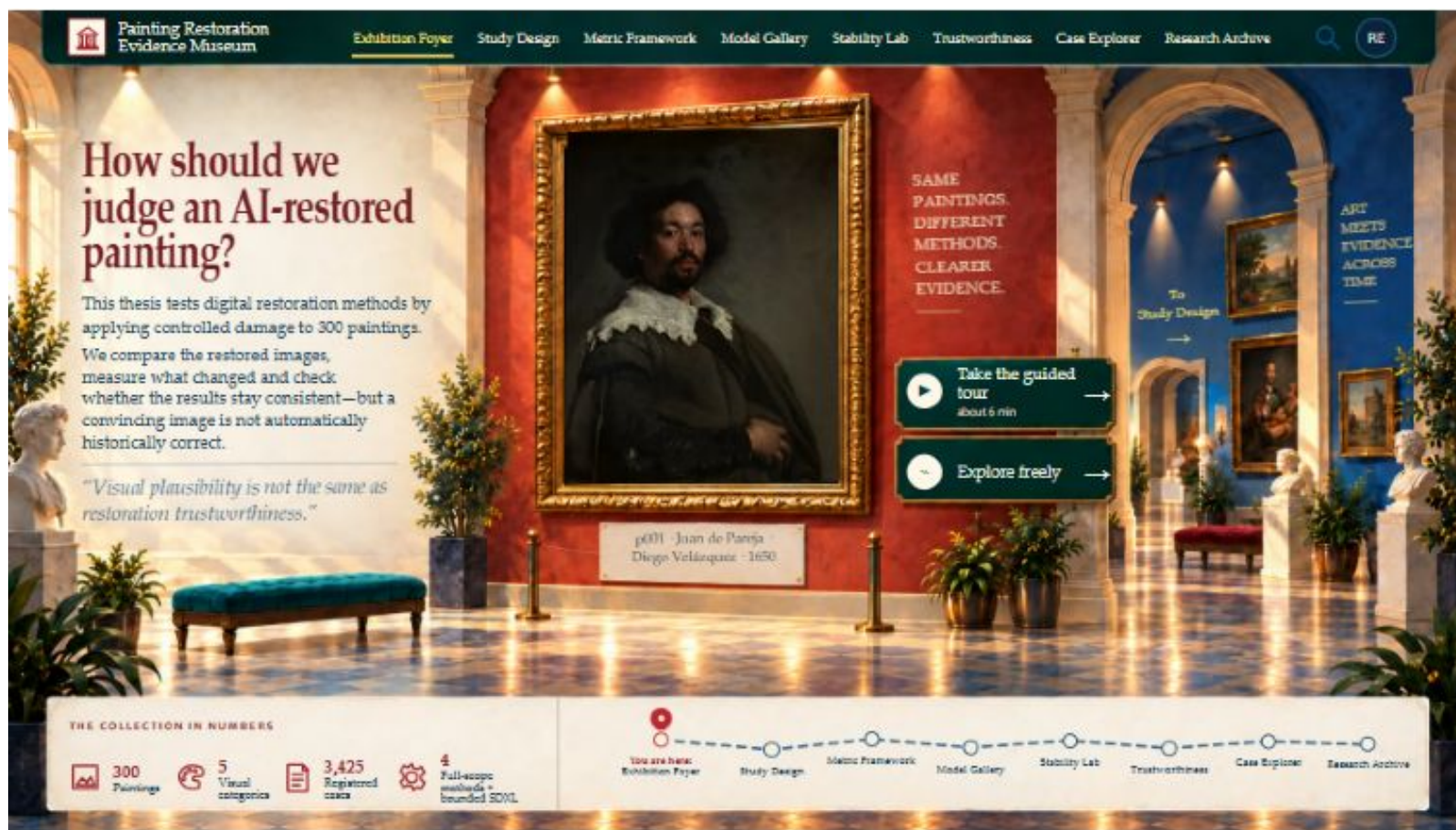


Paintings are the independent units in the grouped analysis. Repeated cases, seeds and metric rows remain linked to their parent painting.

Evidence trail N33 t02/t03; N26 grouped statistics; N36 package README and reproducibility appendix.

Exhibition Foyer

Open with the problem, not the leaderboard.



LOOK LEFT

The central research question introduces evaluation rather than image generation.

LOOK CENTRE

The intact portrait anchors the known-reference experiment.

LOOK RIGHT

Guided tour gives an approximately six-minute route; Explore freely offers question-led entry points.

Say: A restoration can look plausible while changing the painting. This study evaluates what was recovered, what changed, and how strong the supporting evidence is.

Key reading: Start with 300 paintings, four full-coverage methods and controlled damage. The dashboard presents saved evidence; it does not run the models.

Introduce the museum in thirty seconds



ENLARGED LIVE SCREENSHOT DETAIL

SAY THIS

"We compare restored images against known references, then check both quality and stability. A visually attractive result is only the start of the evaluation."

THE OPENING STATEMENT

Point to the question on the left and the reference portrait in the centre. Explain that an intact image is deliberately damaged, restored and compared with its known original. Then choose the guided route or use the top navigation.

Opening claim	Concrete support
Controlled study	300 paintings; 60 per visual category
Shared comparison	4 full-coverage restoration methods
Designed tests	3,425 registered cases
Evidence first	Images, region metrics, maps, flags and records

CONTINUE TO STUDY DESIGN

Use the top navigation. Explain the population and the difference between a registered condition and an eligible restoration case before discussing model performance.

Evidence trail README study scope; N33 tables t01-t03; deployed Exhibition Foyer.

Study Design

Establish what was compared and which cases count.



FOUR BRANCHES

Core test, damage size, mask placement and other changes separate different sources of variation.

KNOWN REFERENCE

The clean painting, altered input and mask define each test.

ELIGIBILITY

Some registered degradation conditions are not inpainting tasks and do not enter the four-method comparison.

Say: The experimental branches change damage size, mask geometry or synthetic degradation while retaining the case identities needed for fair comparisons.

Key reading: 3,425 registered cases are not 3,425 restored cases. The eligible set is 2,620: 2,320 nonzero-damage cases and 300 identity controls.

Keep the denominators straight



ENLARGED LIVE SCREENSHOT DETAIL

FULL COMPARISON

2,620 eligible cases x 4 methods = 10,480 matched primary candidates. The 300 undamaged controls test identity preservation; overall nonzero-damage quality summaries exclude them.

EXPERIMENTAL SCOPE

The room lets the supervisor connect a visible test configuration to its painting, damage and eligibility. Use the lower branch summary to distinguish full-population canonical tests from the 35-painting extensions.

Branch	Cases	Design
Canonical	1,500	300 paintings x 5 conditions
Damage size	245	35 x 7 levels (2%-20%)
Mask placement	525	35 x 15 conditions
Synthetic degradation	1,155	35 x 33 conditions

WHY 13,879 CANDIDATES?

The retained catalogue also contains repeated seeds, prompt variants and bounded SDXL evidence. It is a different population from the matched primary comparison.

Evidence trail N33 tables t02_dataset_design_scope and t03_model_coverage; README population tables.

Metric Framework

Explain what each measurement sees.



EVIDENCE LENSES

Select a question such as spatial improvement, colour or semantic structure.

REGION MATTERS

The mask, crop and boundary answer different questions.

READ THE LEGEND

Signed improvement uses blue for improvement and red for worsening; pale regions indicate little change.

Say: We inspect the missing region, its context and its boundary separately, then compare pixel, perceptual, feature, colour, texture and structural evidence.

Key reading: Telea leads crop SSIM, but LaMa leads perceptual, colour, feature and local-consistency anchors. Different measurements expose different restoration behaviours.

Show why SSIM alone changes the answer



ENLARGED LIVE SCREENSHOT DETAIL

FROM A MAP TO A MEASUREMENT

Use the central inspection area to explain where an effect occurs, and the saved-measurement view to explain its magnitude. A complete canvas can dilute the repaired region with unchanged background; the crop and mask keep the repair in focus.

Mean quality anchor	LaMa	Telea
Crop SSIM (higher better)	0.8577	0.8632
Crop LPIPS (lower better)	0.1292	0.2007
Masked CIEDE2000 (lower)	6.88	7.77
Crop DINOv2 similarity (higher)	0.8712	0.6870

EXPLAIN THE DISAGREEMENT

Telea leads structural similarity in the crop. LaMa more closely matches the reference on perceptual distance, colour and DINOv2 feature similarity. These are distinct observations about the same benchmark.

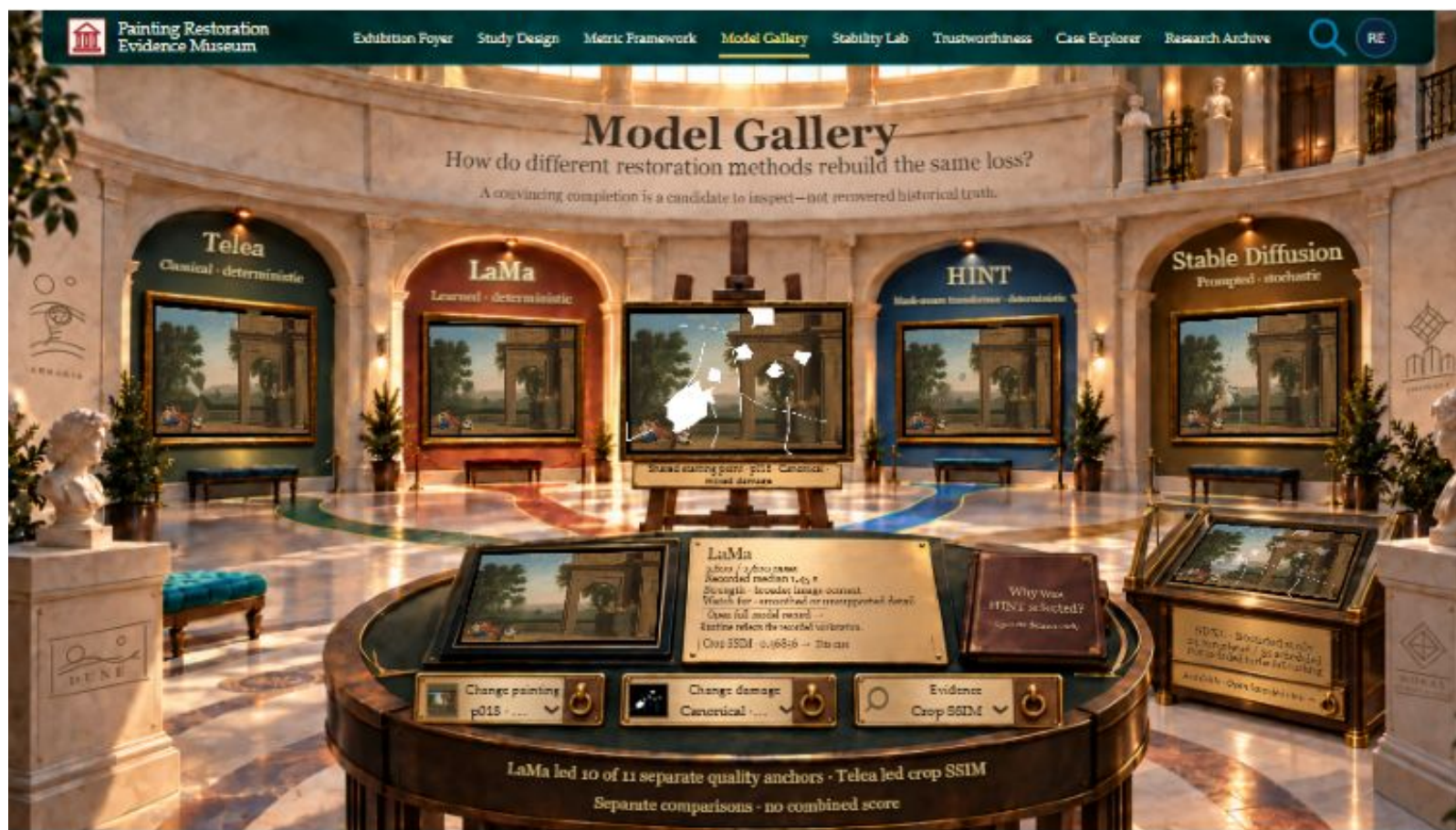
USEFUL NEXT VIEW

Open Model Gallery for the four-method comparison, or Case Explorer to inspect the exact image and saved rows behind one result. Map colours are meaningful only with their current legend.

Evidence trail N33 t04_quality_anchor_summary (overall nonzero-damage means); N28 metric/region ablation.

Model Gallery

Compare methods on the same damaged painting.



UPPER PORTRAITS

Telea, LaMa, HINT and Stable Diffusion share the selected input.

METHOD RECORD

Coverage, runtime and evidence belong to the selected method and case.

BOUNDED BRANCH

SDXL has 24 completed results from 35 scheduled cases and is shown separately.

Say: LaMa leads ten of eleven separate quality anchors in Controlled-300. Telea leads crop SSIM. The gallery connects those aggregate results to individual reconstructions.

Key reading: Overall masked MAE: LaMa 14.83, Telea 17.57, HINT 19.16 and Stable Diffusion 30.61. Lower is better.

Present the comparison without losing its detail



FOUR METHODS, ONE SHARED INPUT

Keep the painting and damage fixed when changing the selected method. Read the visible case metric as a case result, then use the table to explain the overall nonzero-damage comparison.

Method	Masked MAE	Crop LPIPS
LaMa	14.83	0.1292
Telea	17.57	0.2007
HINT	19.16	0.1748
Stable Diffusion	30.61	0.1972

ENLARGED LIVE SCREENSHOT DETAIL

ANSWER THE HINT QUESTION

The separate D01 selection compared HINT and MAT on 12 paired cases. HINT led 96 of 108 case-level anchors, MAT led 6, and 6 tied. HINT was then evaluated across the full eligible set.

EXPLAIN CONSISTENCY

Canonical painting-level comparisons distinguish the four methods on all 11 anchors: 300 paintings, adjusted q below 10^{-134} , and Kendall W of 0.69-0.82. Case-level outcomes can still differ.

Evidence trail N33 t04 and t10; N21 comparison; N26 grouped statistics; D01/N37 method-selection report.

Stability Lab

Ask what happens when the test changes.



VISIBLE TRAJECTORY

Default: p018, LaMa, spatial masked error across seven damage sizes.

RECORDED PAIR

The input and restoration correspond to the selected point.

SEED AVAILABILITY

Repeated-seed evidence is for Stable Diffusion, not deterministic LaMa, Telea or HINT.

Say: Restoration quality and robustness are separate questions. This room changes damage size, mask placement, eligible degradation or the seed while retaining the recorded evidence.

Key reading: The visible p018 LaMa example rises from 4.13 masked error at 2% damage to 39.33 at 20%. Use it as a case example, then give the 35-painting result.

Separate sensitivity from seed disagreement



ENLARGED LIVE SCREENSHOT DETAIL

READ THE CURVE AND THE SELECTED CASE TOGETHER

The curve is a recorded trajectory, not a predicted threshold. Damage-size comparisons span 35 paintings. For seed stability, compare the four outputs of one fixed case and prompt, then inspect variability and pairwise-distance records.

Evidence	LaMa	Stable Diffusion
MAE slope per +10 percentage points	3.43	5.20
Mask-placement MAE dispersion (MAD)	0.92	2.35
Mask-placement LPIPS dispersion (MAD)	0.0037	0.0159

THE REPEATED-SEED RESULT

1,025 four-seed groups provide 6,150 within-group seed pairs. Canonical groups are prompt-specific: 480 generic plus 300 scratch-aware. The damage-size extension adds 245 groups.

RELATIONSHIP TO QUALITY

Across 300 paintings, generic-prompt RGB variability relates to masked MAE (ρ 0.513); LPIPS disagreement relates to reference LPIPS error (ρ 0.586). Scratch-aware associations are 0.622 and 0.713.

Evidence trail N33 t06-t09; N26 statistical_results: uncertainty_performance_association; live p018 trajectory.

Trustworthiness

Trace a review flag back to the measured evidence.



CANDIDATE FIRST

Default: p002, Stable Diffusion, large loss, seed 2026.

THRESHOLD CONTEXT

The selected texture indicator is compared with its own recorded reference stratum.

RECOMMENDATION LAST

Inspect the evidence chain before interpreting the displayed recommendation.

Say: The six-step chain connects an exact candidate, fair comparison population, threshold, failure category, review flags and recorded recommendation.

Key reading: The visible texture-error value is 10.0533, above the warning threshold 1.7924 and critical threshold 4.8924. That is the traceable reason for this displayed flag.

Walk through one flag, step by step



ENLARGED LIVE SCREENSHOT DETAIL

READ MISSING EVIDENCE EXPLICITLY

The default record marks colour as unresolved. An insufficient-evidence status is not a low error, a zero, or a passed check. The flag chain preserves this distinction.

THE EVIDENCE-TO-RECOMMENDATION CHAIN

Follow the numbered route from candidate evidence to recommendation. Comparison peers, experiment, indicator, region and statistic explain the threshold population. The room also exposes policy sensitivity and all eleven flag types.

Visible record	Value / reading
Local texture error (95th percentile)	10.0533
Warning reference	1.7924
Critical reference	4.8924
Selected failure category	Texture smoothing - critical
Recorded recommendation	Unstable candidate - inspect manually

PORTRAIT SIDE ROOM

The D02 doorway opens a separate hand-anatomy study: 45 matched cases across 20 paintings, plus blind review and rendered-lightness analysis. See pages 21-22.

Evidence trail Deployed p002 Trustworthiness record; N27 failure flags; N30 review recommendations; separate D02.

Case Explorer

Follow one exact restoration from image to source row.



FIVE FRAMES

Clean, damaged, mask, restoration and selected evidence layer.

METRIC LEDGER

Damaged and restored values are shown with their direction of improvement.

EXACT IDENTITY

Reports and provenance preserve the selected candidate rather than borrowing another result.

Say: This is the close-reading room: choose the painting, condition, method and candidate, then connect the visible repair to its maps, metrics and reports.

Key reading: Default p018 / mixed damage / LaMa: colour error falls from 52.963 to 4.066; crop LPIPS falls from 0.410 to 0.019; crop SSIM rises from 0.883 to 0.969.

Read one case as an evidence chain



THE METRIC SNAPSHOT AND ITS NEIGHBOURING RECORDS

Read the image comparison first, then the three-row metric snapshot. Use the ledger for exact saved rows and Reports & provenance for the source. The screenshot values are rounded for display; the stored rows retain precision.

Measure / region	Damaged	Restored	Change
Colour / masked	52.963	4.066	48.897 lower
LPIPS / mask crop	0.410	0.019	0.391 lower
SSIM / mask crop	0.883	0.969	0.086 higher

ENLARGED LIVE SCREENSHOT DETAIL

DO NOT CONFUSE ADJACENT EXAMPLES

The retrieval shelf is explicitly a stored p018 dirt/dust HINT example, not the current mixed-damage LaMa case. "What could change?" opens selected recorded comparisons, not new model runs.

AVAILABILITY IS PART OF THE ANSWER

Seed uncertainty is not applicable to the selected deterministic method. The review card still reports one missing required colour indicator. A visible metric and a complete flag policy are different requirements.

Evidence trail Deployed candidate__lama__canonical__p018__mixed_damage__c00; N32 reports; exact saved metric rows.

Research Archive

Close the demonstration with provenance.



SHELVES

Search paintings, cases, candidates, reports and models.

OPEN NOTEBOOK

N33 final evaluation: 24 figures, 49 claims and 18 recorded limitations.

PUBLICATION DESK

GitHub holds code/compact records; HF holds candidates and diagnostic bundles.

Say: The archive lets a reader trace a claim to its report, run manifest and checksum, and locate the corresponding compact or bulk evidence.

Key reading: The 536/536 seal belongs to N33 final-report validation. N36 is the later review-package closeout: 859 passed checks, 16 disclosed warning nonpasses and zero blockers.

Distinguish the study report from delivery closeout



THE FINAL EVALUATION NOTEBOOK ON THE DESK

Open report, Trace sources and Verify checksum connect the visual notebook to saved records. Source Ledger, Run Manifests, Model Cards and Checksum Index offer different routes to the same provenance chain.

Record	What it contains
N32 reports	30 case reports + 300 painting reports + index = 331
N33 final evaluation	24 figures; 49 claims; 18 limitations; 536/536 checks
N36 review package	123 package files within 134-file delivery
N36 evidence index	693 artifact rows; 38 upstream manifests

ENLARGED LIVE SCREENSHOT DETAIL

HAND OVER THE CORRECT PACKAGE

Share the complete N36 output directory, not package/ alone. It includes six scientific HTML reports, 24 figures and five model cards. The reproducibility appendix explains external evidence and environments.

Evidence trail Deployed Archive; N32/N33 manifests; N36 supervisor summary, reproducibility appendix and closeout checks.

Focused Portrait Review

Open the portrait doorway in Trustworthiness. This is a separate audit, not part of the main ranking.



SCREENING

60 portraits and 91 reviewed regions lead to 45 matched hand cases across 20 paintings.

MATCHED CONTROLS

Compare damaged hands with the recorded control region, not with unrelated paintings.

BLIND REVIEW

32 selected reviews hide model names. The anatomy observations are recorded separately.

Say: All 12 method-by-metric estimates show worse performance for damaged hands; 10 remain supported after correction. No method minimizes every hand penalty.

Read carefully: Rendered lightness describes image appearance, not race, identity or ethnicity.

Anatomy evidence and rendered lightness



ENLARGED LIVE SCREENSHOT DETAIL

EXPLAIN THE SELECTED REVIEW

Use the clean, damaged, restored, difference and matched-control crops to describe the anatomy. The default R005 record shows 1,196 damaged hand pixels and 26.28% affected. It is one selected example, not the whole audit.

KEEP THE LIGHTNESS CONCLUSION PRECISE

The default chroma-error view shows three negative associations and one interval crossing zero. Other diagnostics differ, so there is no consistent overall lightness association across all outcomes. Do not turn image lightness into a demographic label.

THE HAND-STUDY CONCLUSION

The matched study asks whether damage affecting hands is harder to reconstruct. The separate lightness analysis asks a different, exploratory question. Keep the two conclusions apart.

Saved result	Reading
12 / 12	Estimated hand penalties are adverse
10 / 12	Supported after multiplicity correction
25 / 32	Selected blind reviews record visible anatomy failure
3 / 4	Negative chroma-error/lightness associations in the visible default view

Evidence trail Deployed Focused Portrait Review and recorded D02 hand/lightness analyses; N36 supervisor synthesis.

Five distinctions that keep the discussion clear

Distinction	What to say
Case result vs aggregate result	p018 is a worked example. The headline comparison uses all eligible nonzero cases; canonical statistical tests use painting-level aggregation.
Quality vs sensitivity	A low error and a small response to a perturbation are different outcomes. Stable Diffusion can show a shallow LPIPS damage-size slope while retaining higher baseline error.
Seed variability vs confidence	Pairwise distances measure disagreement among outputs. Their associations with reference error are empirical; no calibrated probability of correctness is claimed.
Flag vs expert verdict	The eleven computational flags compare specific evidence with recorded policies. Missing evidence remains unresolved. A flag directs inspection; it does not substitute for a conservator.
Validation vs deployment guarantee	N33 report checks, N35 dashboard checks and N36 packaging checks cover different artifacts. A successful screenshot capture is not a full browser or infrastructure regression test.

CLOSE WITH THE CONTRIBUTION

The study delivers a reproducible, region-aware evaluation framework; a controlled four-method benchmark; robustness and stochastic-stability evidence; and a dashboard that lets a reviewer trace conclusions back to saved artifacts.

Evidence trail N33 limitations register; N35 qualifications; N36 limitations/deviations and reproducibility appendix.

Where to find the evidence after the meeting

[Live eight-room dashboard](#)

[Final evaluation report \(N33\)](#)

[Final quantitative tables \(N33\)](#)

[Grouped statistical analysis \(N26\)](#)

[Saved statistical rows \(N26\)](#)

[Metric and region-policy ablation \(N28\)](#)

[Supervisor synthesis \(N36\)](#)

[Reproducibility appendix \(N36\)](#)

HTML reports: GitHub shows their source. Download and open locally, or access them through the dashboard, to see the rendered report.

Delivery status: N36 is complete: 859 passes, 16 disclosed warning nonpasses, zero blockers. Its 134-file output includes a 123-file review package. Preserve the complete output directory when sharing.

This guide updates the pilot walkthrough for Controlled-300. It introduces no new experiments and does not change the frozen dashboard or scientific records.

CAPTURE RECORD

Nine live room captures were made on 02 October 2026: all eight top-level tabs plus Focused Portrait Review. Full-room images preserve the loaded default state; details are crops of those same screenshots, not mockups or rebuilt UI.

VERSION RECORD

Local documentation baseline: commit 5ede9ba4. Scientific package: N36 run_e3133aca48134968896bcd1b97af72c4. This does not independently attest the deployed server revision.